

Data Analysis Plan

Voice-Based Screening of Parkinson's Disease Using Clustered Acoustic Features

Background and Goal

Parkinson's disease (PD) is a progressive neurodegenerative disorder characterized by motor and non-motor symptoms, many of which manifest as subtle but measurable changes in speech. (Cao et al., 2025; Ma et al., 2020) Prior work has demonstrated that acoustic features derived from voice recordings can capture aspects of hypophonia, reduced articulatory precision, altered prosody, and voice instability associated with PD. (Shen et al., 2025; Yang et al., 2020) However, translating these findings into clinically deployable screening tools remains challenging due to high-dimensional feature spaces, redundancy among acoustic measures, inter-recording variability, and limited generalizability at the patient level. (Naeem et al., 2025; Wayland et al., 2025)

Recent large-scale efforts such as the Bridge2AI Voice dataset provide an unprecedented opportunity to systematically study voice biomarkers using standardized speech tasks and rich metadata across diverse participants. (Bensoussan et al., 2025; *Datasets – BRIDGE2AI*, 2025) Preliminary analyses suggest that Parkinson's-related voice differences are subtle, distributed across multiple weak acoustic signals, and not dominated by a small number of highly discriminative features. (Rusz et al., 2024; Wayland et al., 2025) This motivates a modeling strategy that emphasizes structured dimensionality reduction, feature aggregation, and patient-level inference rather than reliance on raw high-dimensional features. (Tsanas & Arora, 2022)

The overall goal of this project is to develop and evaluate a reproducible, voice-only machine learning pipeline for Parkinson's disease screening that is suitable for clinical translation. (Bensoussan et al., 2025; Xu et al., 2025) The primary goal is to develop a reproducible, voice-only machine learning pipeline for patient-level Parkinson's screening with strong generalizability. Secondary goals are to (i) reduce redundancy by learning compact cluster-aggregated acoustic representations, (ii) characterize the modular structure of the acoustic feature space to guide dimensionality reduction, and (iii) enhance interpretability by linking predictions to coherent acoustic feature families.. (Tsanas & Arora, 2022; Wayland et al., 2025)

Tasks

The primary tasks of this analysis are:

- To characterize the correlation structure of static acoustic voice features and identify coherent feature families using hierarchical clustering.
- To construct cluster-aggregated acoustic representations by summarizing correlated feature groups using simple, interpretable aggregation rules (e.g., mean, median).
- To compare the performance of baseline models trained on the full feature set with models trained on progressively reduced cluster-level representations.

- To evaluate model performance at both the recording level and the patient level, with an emphasis on patient-level discrimination as a more clinically meaningful outcome.
 - Recording level: Each speech sample treated independently.
 - Patient level: Aggregation of multiple recordings per participant using summary statistics (mean or median), reflecting a more realistic clinical screening scenario.
- To assess model stability and generalization across multiple classical machine learning classifiers without incorporating demographic or clinical phenotype variables.

The analysis will explicitly focus on a **voice-only approach**, excluding phenotype variables, to ensure fairness, portability, and suitability for real-world deployment scenarios.

Dataset

The analysis will use data from the **Bridge2AI Voice Dataset (PhysioNet v2.0.1)**, which contains over 19,000 voice recordings collected using standardized speech tasks across a diverse participant population. (Bensoussan et al., 2025; *Datasets – BRIDGE2AI*, 2025)

For this project, we will focus on a Parkinson’s disease subset consisting of:

- **Parkinson’s disease cases:** 61 participants
- **True controls:** 64 participants

Each participant contributes multiple speech recordings across standardized tasks. For each recording, **131 static acoustic features** are available, summarizing pitch, loudness, spectral, cepstral, jitter, shimmer, and temporal speech characteristics.

Recordings will be analyzed at two levels:

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Methods

1. **Exploratory Feature Structure Analysis**
 - Compute pairwise correlations among acoustic features.
 - Perform hierarchical clustering using a distance metric defined as $(1 - |r|)$.
 - Identify modular structure and determine meaningful cluster cuts representing coherent acoustic feature families.
2. **Dimensionality Reduction and Visualization**
 - Apply linear (PCA) and nonlinear (UMAP) embeddings to assess variance distribution and label separability.
 - Examine whether Parkinson’s disease signal emerges globally or remains locally distributed across the feature space.
3. **Cluster-Aggregated Feature Construction**
 - Replace the original 131 features with (K) cluster-level features by aggregating features within each cluster.
 - Evaluate multiple values of (K) (e.g., 5–60 clusters).

- Compare aggregation strategies (mean, median, PCA-based summaries).
- 4. **Model Training and Evaluation**
 - Train baseline classifiers on the original feature space and cluster-aggregated feature spaces, including:
 - Logistic Regression
 - Random Forest
 - Gradient Boosting (XGBoost)
 - Regularized Residual Multilayer Perceptron (RR-MLP)
 - Perform cross-validation with patient-level separation to prevent data leakage.
 - Evaluate both recording-level and patient-level performance.
- 5. **Interpretability Analysis**
 - Use SHAP (Shapley Additive Explanations) to study how models utilize different acoustic clusters.
 - Compare correlation-based feature groupings with model-usage-based feature relationships.

Deliverables

- Fully reproducible and well-documented computational pipelines implemented in Python.
- Cross-validated performance results comparing:
 - Baseline models using all acoustic features.
 - Reduced models using cluster-aggregated acoustic representations.
- Patient-level and recording-level evaluation summaries.
- Visualizations of feature clustering, embeddings (PCA/UMAP), and SHAP-based interpretability.
- A concise analytical report suitable for course submission and future manuscript preparation.

Metrics

Model performance will be evaluated using:

- **Primary metrics**
 - ROC-AUC
 - F1 score
- **Secondary metrics**
 - Accuracy
 - Precision
 - Recall
- Primary metric (screening discrimination):
 - ROC-AUC (patient-level and recording-level)
- Secondary performance metric:
 - F1 score (when using a fixed threshold)
- Understanding / representation quality metrics (aligned with the goal of structured reduction):
 - Δ ROC-AUC relative to baseline across K clusters

- Compression ratio (131/K) and selection of the Pareto-optimal K
- Cross-validation stability: SD/CI of ROC-AUC across folds; stability of cluster importance rankings (Spearman)
- Interpretability at the cluster level: concentration of predictive contribution (e.g., proportion of total |SHAP| explained by top clusters)

Metrics will be reported at both the **recording level** and **patient level**, with emphasis on patient-level discrimination.

References

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Uday Singh

1/20/2026

Xiaoqian Jiang

1/20/2026